ONUR KELEŞ, Ph.D.

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Google Scholar | github.com/onur-keles | For more detail: onur-keles.github.io

Education

Ph.D. | ***Electrical & Electronics Engineering*** — Sep. 2019 – Mar. 2025
Koç University — Istanbul, TUR

- Graduation GPA: 4.00/4.00
- Dissertation: Non-linear Neuron Modeling using Padé Approximants with Applications to Single Image Super-Resolution and Image Compression
- Advisor: Prof. A. Murat Tekalp

M.Sc. | ***Electrical & Electronics Engineering*** — Sep. 2016 – Aug. 2019
Boğaziçi University — Istanbul, TUR

- Graduation GPA: 3.50/4.00
- Thesis: Vision Correction for the Visually Impaired via Digital Image Processing

B.Sc. | ***Electrical & Electronics Engineering*** — Sep. 2010 – June 2016
Boğaziçi University — Istanbul, TUR

- Graduation GPA: 3.24/4.00
- Design Project: Automatic Punctuation Generation for Turkish

Work Experience

Senior AI Research Scientist — Feb. 2024 – June 2026
Codeway Digital Services — Istanbul, TUR

- In a team, improved the naturalness of a face reenactment model outputs. Contributed to product: *FaceDance*.
- Led a four-person team to develop and improve lip synchronization, video dubbing, and talking-head generation solutions by post-training an open-source frontier model.
- Increased head video generation from 90 FPS to 2800 FPS with enhanced fidelity using Gaussian splatting, providing the ability for real-time deployment.

AI Research Scientist — Oct. 2022 – Feb. 2024
Codeway Digital Services — Istanbul, TUR

- Doubled the output resolution of a top-chart mobile application for face image enhancement, improving image clarity. Contributed to product: *Pixelup*.
- Developed a fast, efficient solution for multi-face recognition and separation for identity-specific face generation. Contributed to products: *Pixelup*, *Retake*.

Doctoral Researcher — Sep. 2019 – Mar. 2025
Koç University İş Bankası Artificial Intelligence (KUIS AI) Center — Istanbul, TUR

- Advanced research on non-linear neuron models in super-resolution and image compression, published three peer-reviewed conference papers detailing research outcomes.
- Proposed a novel, inherently non-linear neuron model, applied the model to super-resolution and image compression problems, resulting in a published conference paper and a journal paper.

Software Specialist (Part-time) — Feb. 2016 – June 2016
Speech-Enabled Software Technologies (SESTEK) — Istanbul, TUR

- Initiated development of text-to-speech (TTS) software, and prepared a dictionary, for Chinese language.
- Enhanced automatic punctuation generation for Turkish text produced by speech-to-text (STT) software.

Publications

- **Onur Keleş**, A. Murat Tekalp, "*Padé Neurons for Efficient Neural Models*," in IEEE Transactions on Image Processing, vol. 35, pp. 1508–1520, 2026, DOI: 10.1109/TIP.2026.3653202.
- M. Akın Yılmaz, **Onur Keleş**, A. Murat Tekalp, "*The Practice of Averaging Rate-Distortion Curves over Testsets to Compare Learned Video Codecs Can Cause Misleading Conclusions*," in arXiv preprint, DOI: 10.48550/arXiv.2409.08772.
- **Onur Keleş**, A. Murat Tekalp, "*Paon: A New Neuron Model Using Padé Approximants*," in IEEE International Conference on Image Processing (ICIP), 2024, pp. 207–213, DOI: 10.1109/ICIP51287.2024.10648214 (`https://github.com/onur-keles/Paon`).
- **Onur Keleş**, A. Murat Tekalp, Junaid Malik, and Serkan Kıranyaz, "*Self-Organized Residual Blocks for Image Super-Resolution*," in IEEE International Conference on Image Processing (ICIP), 2021, pp. 589–593, DOI: 10.1109/ICIP42928.2021.9506260.
- M. Akın Yılmaz, **Onur Keleş**, Hilal Güven, A. Murat Tekalp, Junaid Malik, and Serkan Kıranyaz, "*Self-Organized Variational Autoencoders (Self-VAE) for Learned Image Compression*," in IEEE International Conference on Image Processing (ICIP), 2021, pp. 3732–3736, DOI: 10.1109/ICIP42928.2021.9506041.
- **Onur Keleş**, M. Akın Yılmaz, A. Murat Tekalp, Cansu Korkmaz, and Zafer Doğan, "*On the Computation of PSNR for a Set of Images or Video*," in Picture Coding Symposium (PCS), 2021, pp. 1–5, DOI: 10.1109/PCS50896.2021.9477470.
- **Onur Keleş**, and Emin Anarım, "*Adjustment of Digital Screens to Compensate the Eye Refractive Errors via Deconvolution*," in Ninth International Conference on Image Processing Theory, Tools and Applications (IPTA). IEEE, 2019, pp. 1–6, DOI: 10.1109/IPTA.2019.8936098.

Research Interests

New model/learning paradigms · Non-linear neuron models · Image and video super-resolution · Learned image compression · Generative video synthesis (talking-head generation, lip synchronisation) · Gaussian splatting

Research Projects

The Scientific and Technological Research Council of Türkiye (TÜBİTAK) Project Spr. 2021 – Spr. 2024
Koç University İş Bankası Artificial Intelligence (KUIS AI) Center, Koç University

- Project: Explainable Deep Learning Approaches for Image and Video Restoration and Compression (120C156)

The Scientific and Technological Research Council of Türkiye (TÜBİTAK) Project Fall 2019 – Spr. 2021
Koç University İş Bankası Artificial Intelligence (KUIS AI) Center, Koç University

- Project: Image and Video Processing with Deep Learning (217E033)

Boğaziçi University Scientific Research Project Spr. 2019
Boğaziçi University

- Project: Adjustment of Digital Screen to Compensate the Eye Refractive Errors via Image Processing (19A02P1)

Professional Services

Teaching Experience

Prepared problem sets; prepared and graded quizzes and MATLAB coding homeworks.

ELEC201 – Signals and Systems | ***Teaching Assistant*** Spr. 2022
Koç University

ELEC301 – Systems, Control and Communications | ***Teaching Assistant*** Fall 2019 – 2021
Koç University

ELEC303 – Digital Signal Processing | ***Teaching Assistant*** Spr. 2020 – 2021
Koç University

ELEC406/506 – Digital Image and Video Processing | ***Teaching Assistant*** Spr. 2021; Fall 2022, 2024
Koç University

Reviewership

IEEE International Conference on Image Processing (ICIP) 2022 – 2024

Achievements, Honors and Awards

TÜBİTAK Project Scholarship Fall 2019 – Spr.2024
Koç University

Boğaziçi University Scientific Research Project Scholarship Spr. 2019
Boğaziçi University

TÜBİTAK 2210-A National Grade Scholarship Fall 2016 – Fall 2018
Boğaziçi University

Academic Personnel and Graduate Education Entrance Exam (ALES) Sep. 2018
National graduate admissions exam; ranked 41st out of around 120,000 participants in Türkiye

University Placement Exam (YGS & LYS) June 2010
National university placement exam; ranked 485th out of 1.8 million participants in Türkiye

Skills and Abilities

Programming: Python, MATLAB, C/C++, C#
Frameworks: PyTorch, NumPy, OpenCV
Languages: Turkish (native), English (proficient), Chinese (pre-intermediate)